# Neural Networks for Data Science

## Master's Degree in Data Science

# Lecture 0: About the course

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# Before we start

- **Master's Degree in Data Science**, code 10589627, 2nd year, optional group D.
- **Timetable**: Thursdays 09:00–12:00 and Fridays 09:00–11:00, room A5 (Via Ariosto), 24 September – 18 December 2026.
- **Office hours**: by appointment, remotely or in person.

https://www.sscardapane.it/teaching/nnds-2026/

Register for **Google Classroom** from the website: it is mandatory for receiving announcements.

# Exam dates (2027)

- **Sessions 1–2**: 13 January, 3 February.
- **Sessions 3–4**: 16 June, 14 July.
- **Session 5**: 15 September.
- **Extraordinary E1**: 7 April.
- **Extraordinary E2**: 13 October.

# Exam (new this year)

Default exam is an **oral examination** on the course material (2 questions).

A **research project** can replace the oral only when it is agreed in advance and it serves one of two purposes:

- pursuing a mark of **30 e lode**;
- preparing for a follow-up **thesis**.

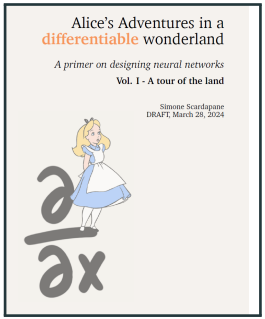
It is intended for interested students and is generally **more demanding** than the oral examination.

# The course in one slide

1. **Fundamentals**
   Arrays, optimization, supervised learning, MLPs, automatic differentiation, and PyTorch (maybe JAX).
2. **Architectures**
   Images, sequences, text, and graphs; convolutions, recurrence, attention, and message passing.
3. **Applications**
   Tokenization, pre- and post-training, self-supervised learning, possibly agents.

# Alice in a differentiable wonderland

Slides are self-contained, but most of the material is expanded in a textbook:

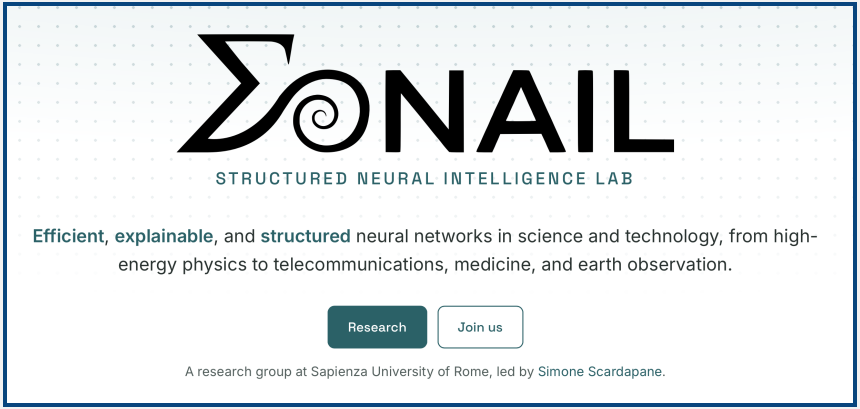



The book is available for free as a PDF or as a printed copy:

`https://sscardapane.it/alice-book/`

You can also find additional lab sessions and chapters (e.g., more on post-training of models and automatic differentiation).

# Structured Neural Intelligence Lab




Explore the group: www.sscardapane.it/snail

# Optional resource: TinyTorch

A hands-on implementation of PyTorch from scratch.

More in-depth than our lab sessions. Let me know if you try them.
Disclaimer: I have not.

`https://mlsysbook.ai/tinytorch/`

